Seeing Clearly or Seeing Through Tinted Glasses:

Accurate and Biased Perceptions of Emotions and Understanding in Close Relationships

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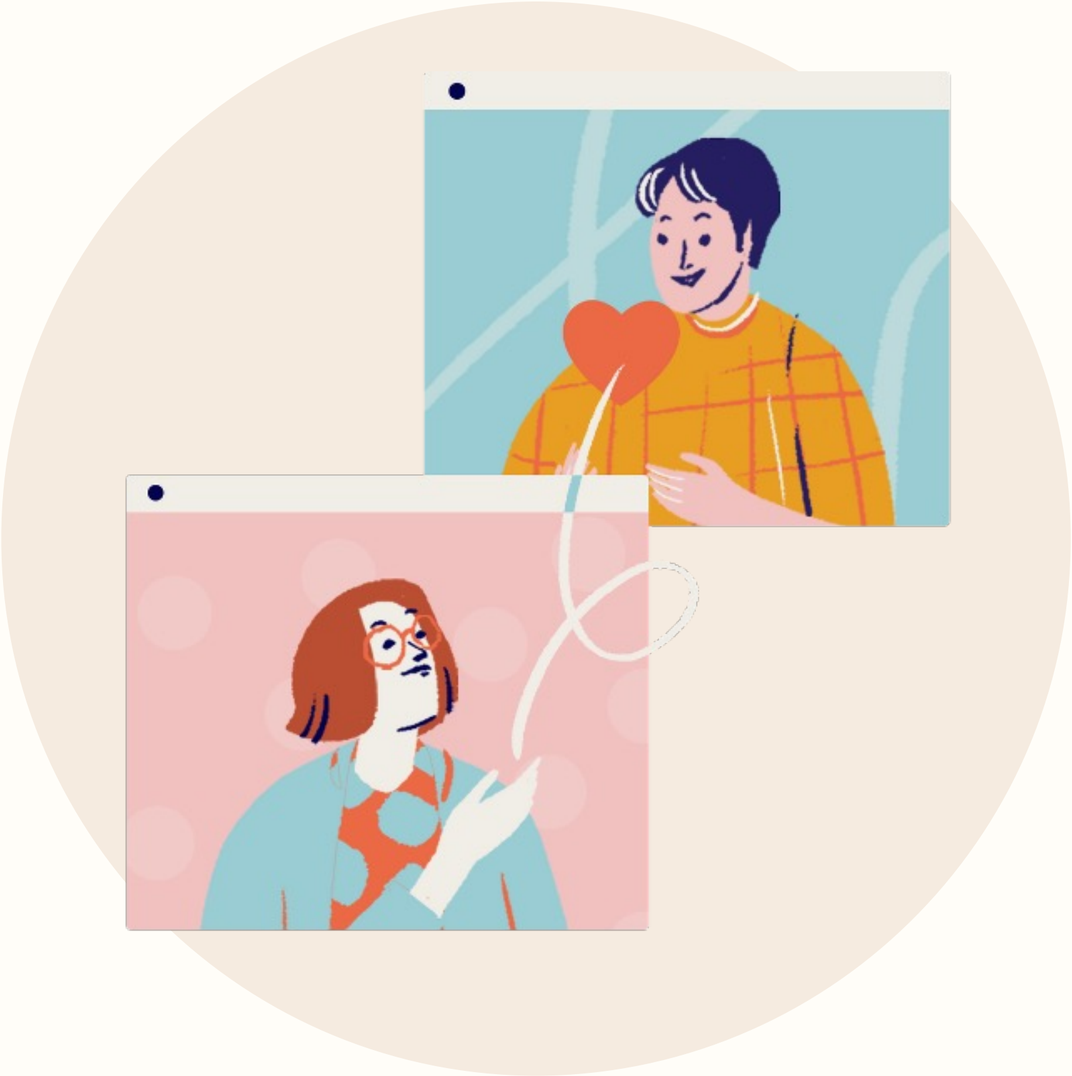
Advisor: Harry T. Reis
May 24, 2024



Theoretical Background

Partners in romantic relationships must
work together to navigate their emotional lives.

Niven et al., 2017; Von Culin et al., 2017; Zaki et al. (2013)

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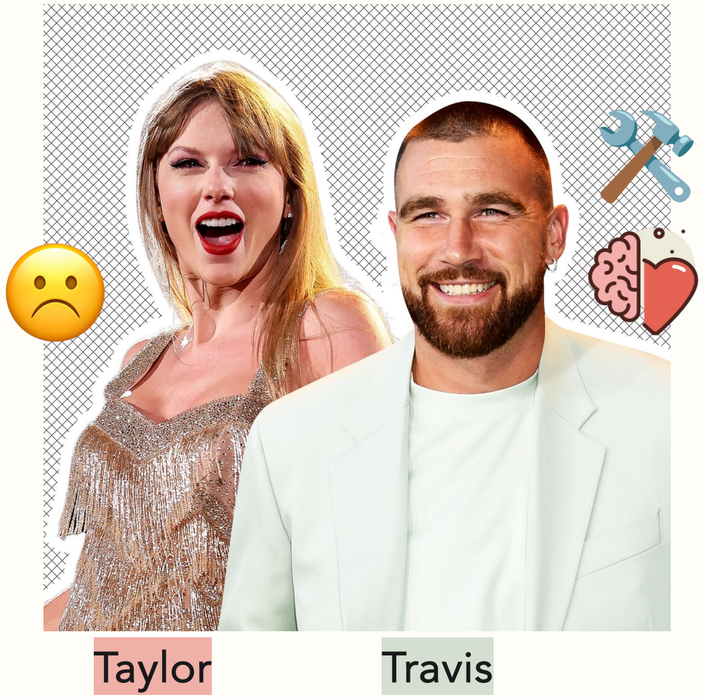


At the end of the day, Taylor shares the emotions that she experienced with Travis.

Travis will try to perceive her emotions.
Travis will attempt to help modify or influence her emotions.
Travis may feel that he understood Taylor.

How does Travis judge his ability to:

- *perceive* Taylor's emotions
- *influence* Taylor's emotions
- *understand* Taylor in a global sense

Is Travis accurate or biased in his perceptions?

Gordon & Chen, 2017; Niven et al., 2017; Reis et al., 2017; Von Culin et al., 2017; Zaki et al. (2013)

Accurate Perceptions (Truth & Bias Framework)

Taylor (Target)

Travis (Agent)

Accurate perceptions:
established when an agent's perceptions of target correspond well with target's self-ratings

Measured as **tracking accuracy** (in Truth & Bias)
correlation or association between agent's ratings of target and target's self-ratings (standard for truth)

Carney et al., 2003; Cohen et al., 2012; Gregory et al., 2020; Hall et al., 2009; O'Brien, 2009; Yoo et al., 2011

Biased Perceptions (Truth & Bias Framework)

Taylor (Target)

Travis (Agent)

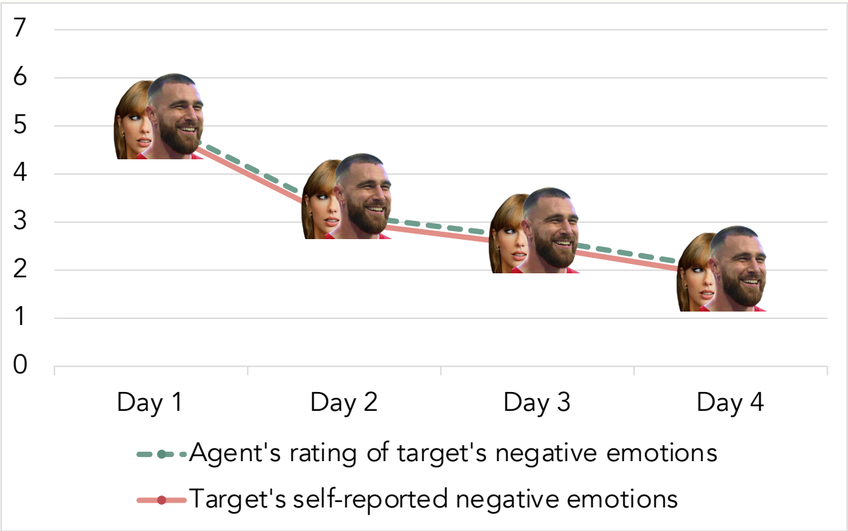
Biased perceptions:
overestimations or underestimations in an agent's perceptions of target, relative to target's self-ratings (standard for truth)

Measured as **directional bias** (in Truth & Bias)

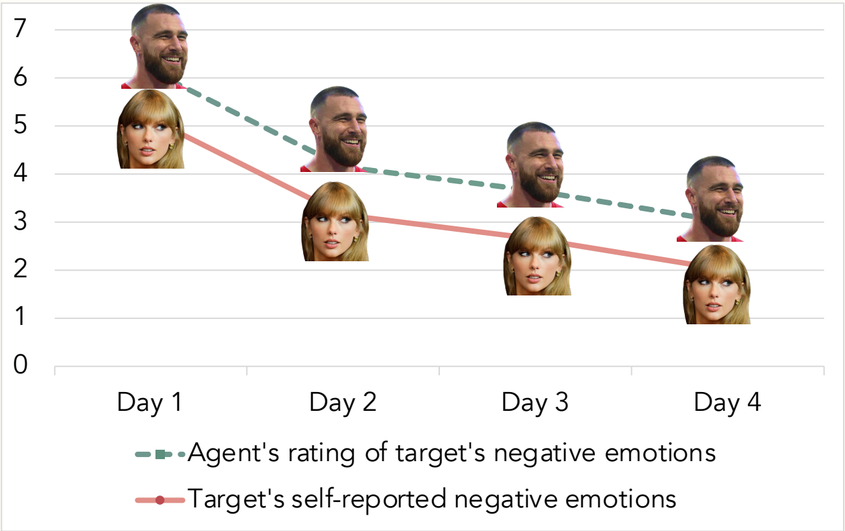
- ***positive directional bias***
  overestimation of positive, underestimation of negative

- ***negative directional bias***
  overestimation of negative, underestimation of positive

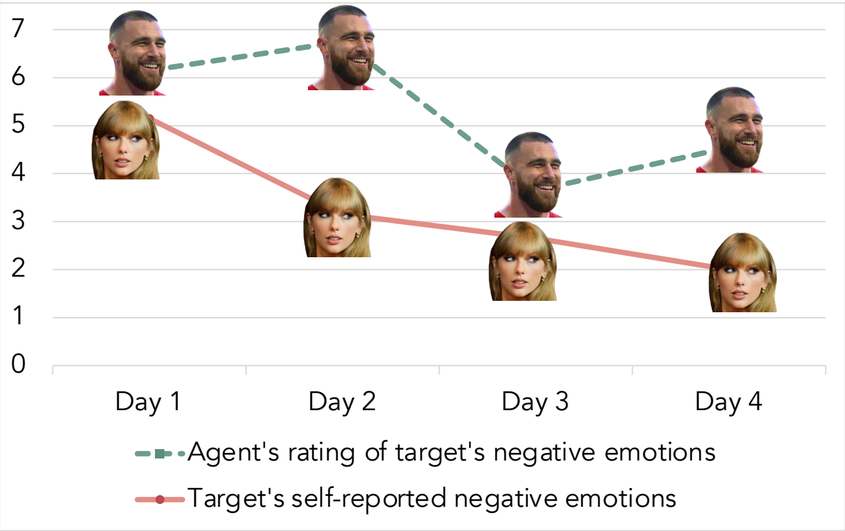
Fletcher & Kerr, 2010; Gagne & Lydon, 2004, La Buda & Gere, 2023



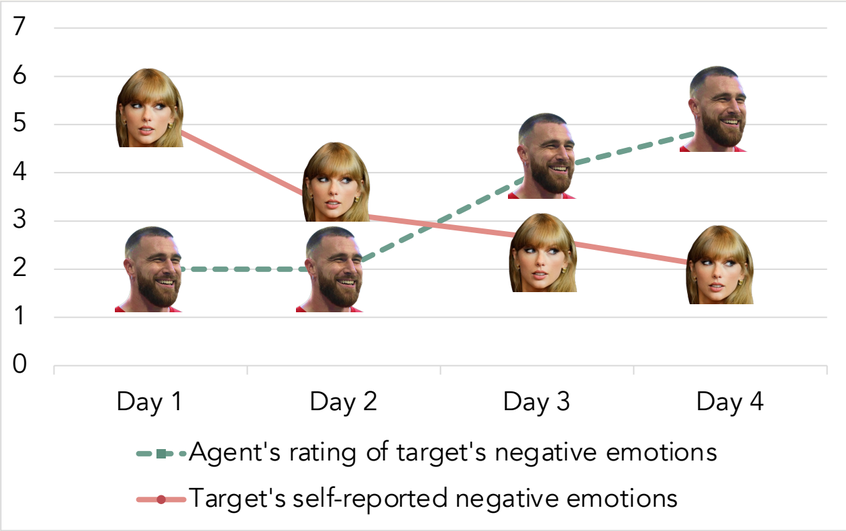
Tracking Accuracy, No Directional Bias



Tracking Accuracy, Positive Directional Bias



No Tracking Accuracy, Positive Directional Bias



No Tracking Accuracy, No Directional Bias

Stern & West, 2018; West & Kenny, 2011

Impact of Accuracy and Bias on Relationship Quality

Accurate perceptions

- Accurately perceiving certain negative and positive emotions have largely been associated with positive relationship outcomes.
- There are certain boundary conditions under which accurate perceptions may be unrelated to or undermine relationship quality.

Biased perceptions

- Few studies examine directional bias within the context of couple's emotional lives.
- Research in other contexts has found the effect of directional bias to be inconsistent.
  - Positive directional bias
  - Negative directional bias
  - No effects or different effects for each partner

Accurate and biased perceptions in the emotional context are at the daily level or concurrent.

Haselton & Buss, 2000; Kim et al., 2023; La Buda & Gere, 2023; Le et al., 2020; Murray et al., 1996; Neff & Karney, 2005; Sened et al. 2017; Simpson et al., 2001, 2003

Present Research

Examine accuracy and bias in 3 constructs of interest relevant to the emotional lives of romantic couples, but not yet explored.

Agent's perception of:

1. Target's daily negative and positive emotions

2. Their influence on target's daily negative and positive emotions

3. Their daily global understanding of target

How do agents' accuracy and bias predict agents' and targets' relationship satisfaction (daily level, longitudinally)?

Taylor (Target)

Travis (Agent)

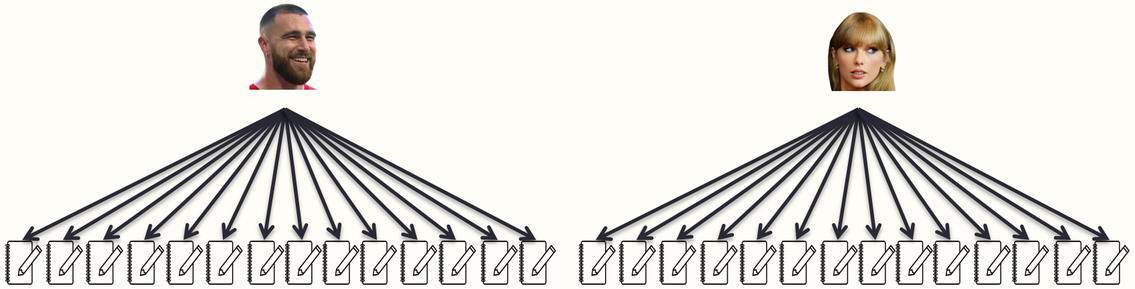
Gregory et al., 2020; Le et al., 2020; Reis et al., 2017; Ruan et al. 2023

Present Research

To examine daily level effects,

Initial Study: Two-week daily diary *(completed in 2019)*

- Both partners of 197 romantic couples reported on perceptions of their emotional lives
    - Cohabitating for at least 1 year
    - $M_{\text{relationship length}} = 9.4$ years
    - 55.1% married,11.2% engaged, 33.8% committed monogamous relationship



To examine longitudinal effects,

Follow-up Study: Follow-up survey with same couples five years later *(to complete in 2024)*

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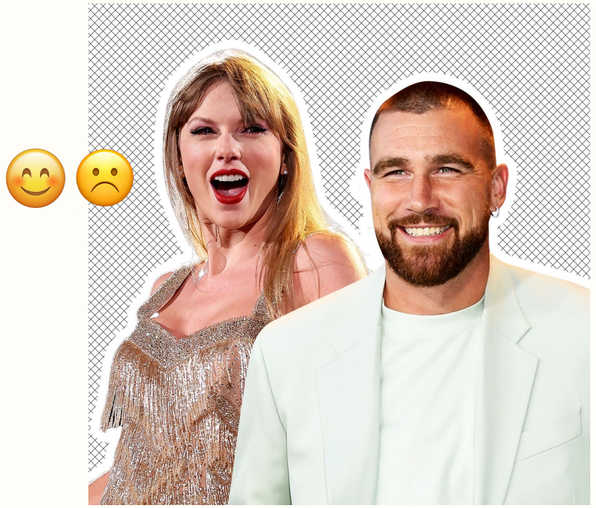
Agents' perception of targets' daily emotions

Agent: Did target experience negative/positive emotions?
Target: Did you experience negative/positive emotions?

Taylor (Target) Travis (Agent)



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Tracking accuracy effect

- Agents will demonstrate tracking accuracy in their perception of targets' daily negative emotions and positive emotions.

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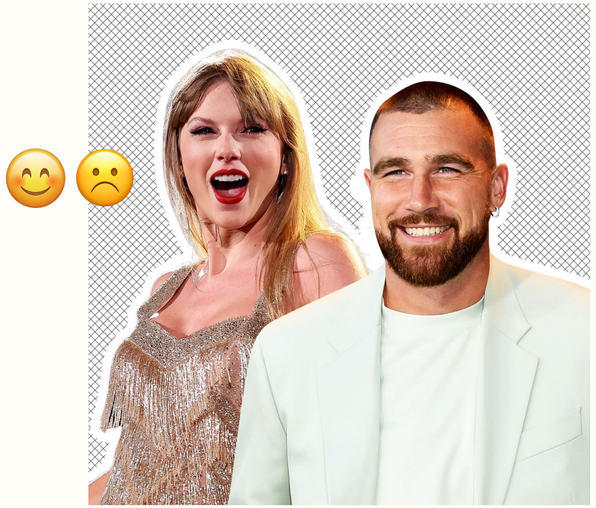

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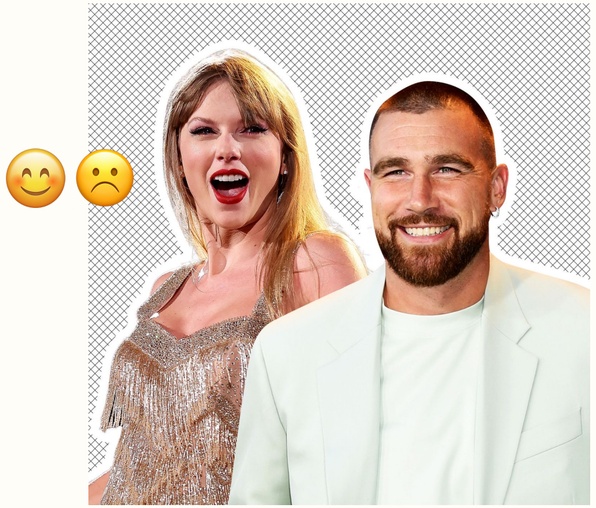

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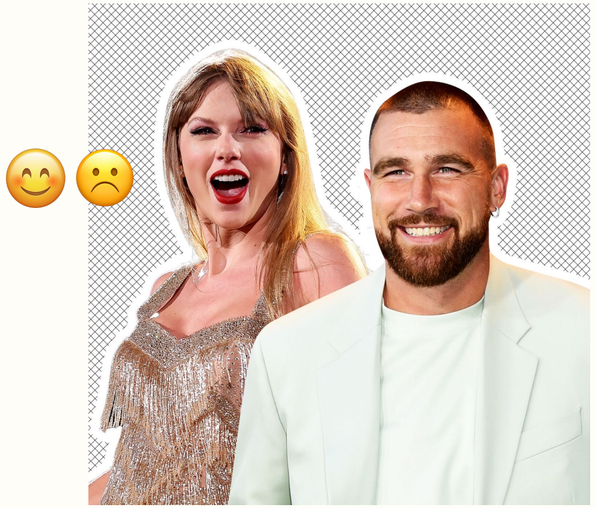

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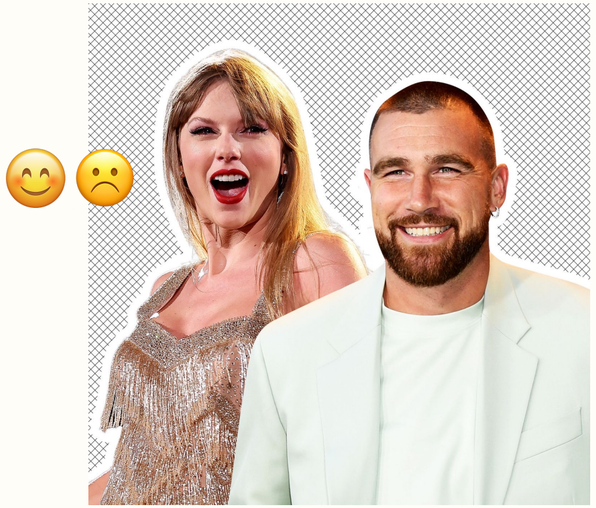

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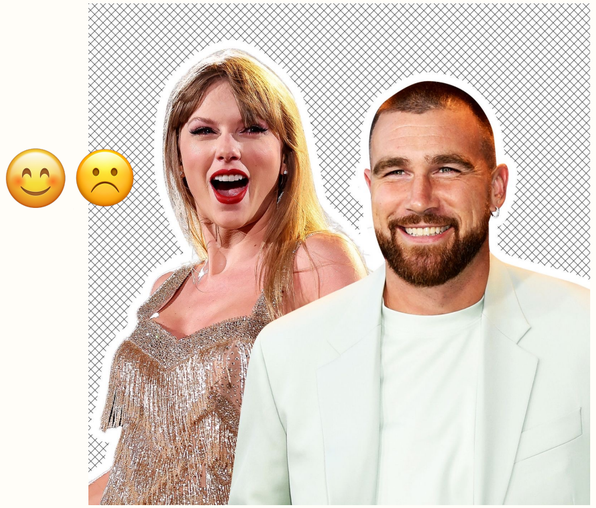

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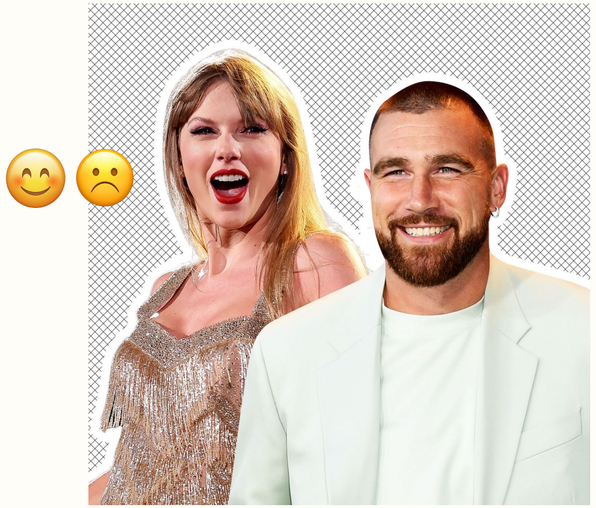

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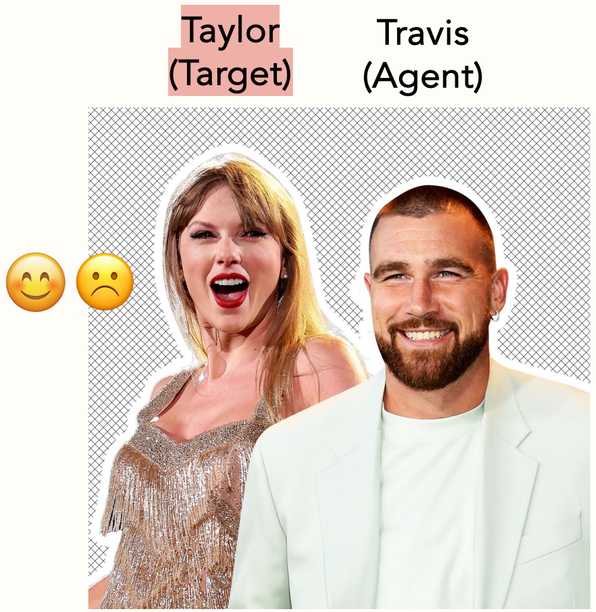

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Hypotheses Set 2:
Agents' perception of their influence on targets' daily emotions

Agent: How effective were you in changing/influencing target's existing negative/positive emotions?
Target: How effective was agent in changing/influencing your existing negative/positive emotions?

Taylor (Target) Travis (Agent)

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Agent: I understood target; I accurately understood target's feelings.
Target: Agent understood me; Agent accurately understood my feelings.

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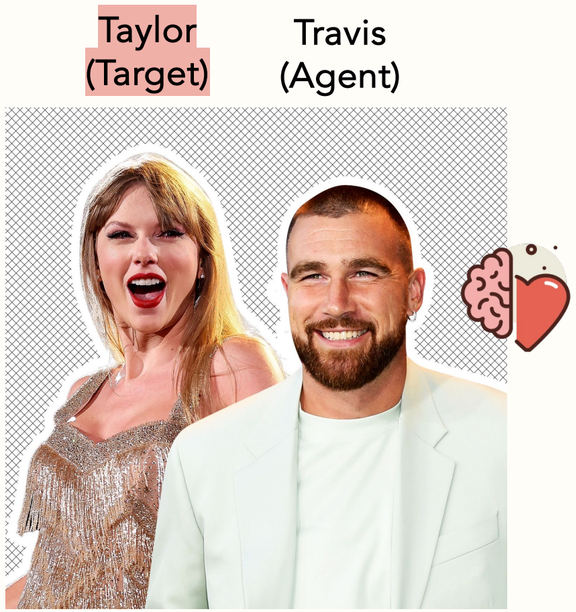

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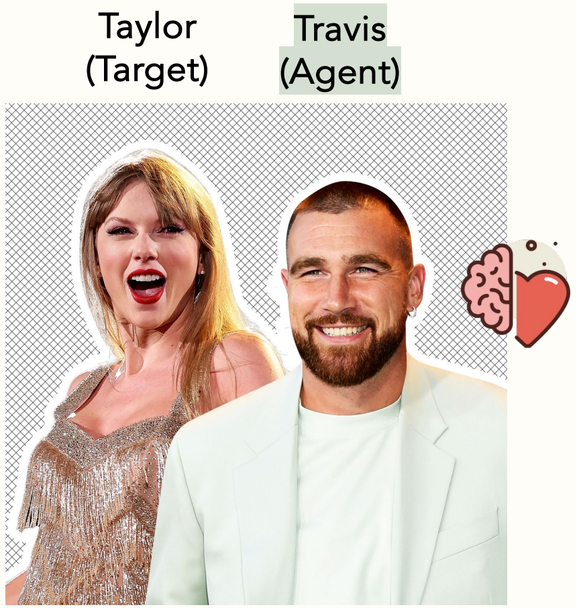

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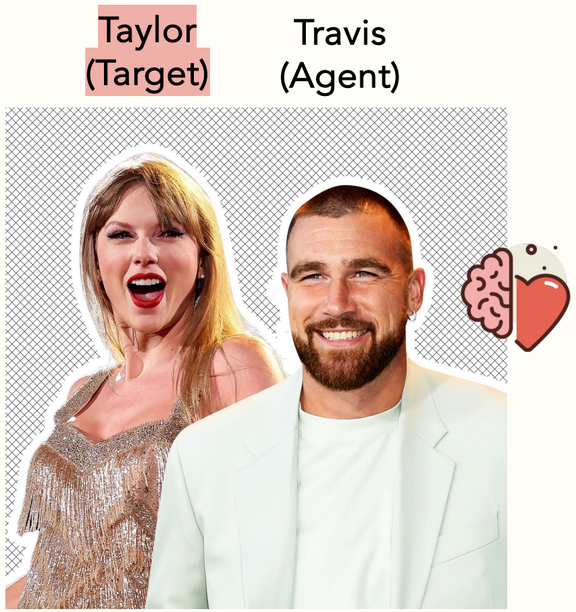

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Follow-up Study: Method

Sample size goal: 100 couples (La Buda & Gere, 2021; Spielmann et al., 2020; West & Kenny, 2011)

Both partners (from initial study) contacted for preliminary survey

- Relationship status
- Cohabitating
- Contact information

Depending on relationship status, sent one of two follow-up surveys

- Together:
  - Full Version (20 minutes, $20): contact once a week for 3 weeks
  - Abbreviated Version (10 minutes, $15): contact once a week for 2 weeks
- Separated:
  - (10 minutes, $15): contact once a week for 5 weeks

Follow-up Study: Measures

If the couple is still together,

- Relationship Satisfaction (Couple Satisfaction Index, 16-item)
- Commitment Level (Rusbult et al.,1998; 7-item)
- All previous perceptions
  - agents' perception of targets' emotions
  - agents' perception of their influence on targets' emotions
  - agents' perception of their understanding of targets' emotions

If the couple has separated, broken-up, or divorced,

- Relationship Change (12-item)
- Separation Feelings (12-item)
- Rekindling Desire (5-item)

Analyses

To test tracking accuracy and directional bias effects

- Using the Truth & Bias framework, agents' perceptions will be centered on targets' actual reports.

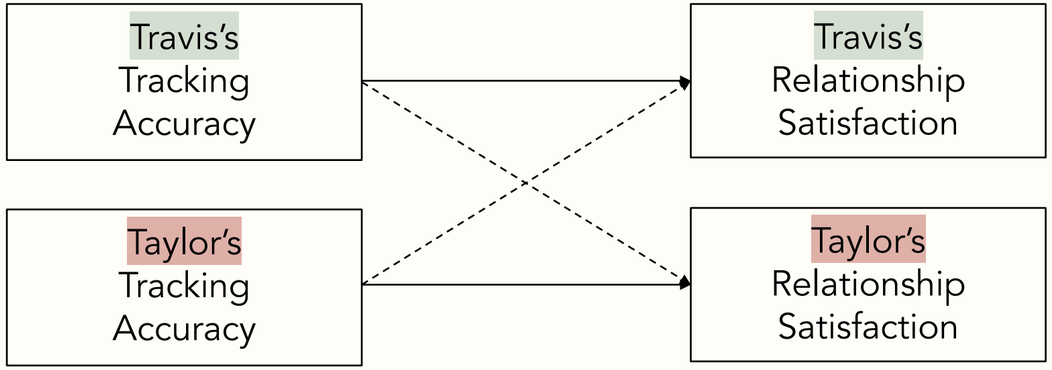
- Using multilevel modeling,
  - Intercepts are the difference between targets' actual reports and agents' perceptions = **directional bias**
  - Slope of targets' actual reports on agents' perceptions = **tracking accuracy**

	Partner 1 Tracking Accuracy	Partner 1 Directional Bias	Partner 2 Tracking Accuracy	Partner 2 Directional Bias
Couple 1	Taylor's TA	Taylor's DB	Travis's TA	Travis's DB
Couple 2	Brittany's TA	Brittany's DB	Patrick's TA	Patrick's DB

Analyses

To test accuracy and bias associations with relationship satisfaction (daily level)

- Using Actor-Partner Interdependence Model (APIM),



APIMPower: 98% power to detect actor effect (β = .20), 52% power to detect partner effect (β = .10)

To test accuracy and bias associations with relationship satisfaction (longitudinal level)

- Using APIM, examine association between **each partner's tracking accuracy** (or **directional bias) and their own and their partner's relationship satisfaction five years later**.

Thank you!

Thoughts, feelings, questions, or comments?